PHY224 - Three-Hole Aluminium Foil Imaging

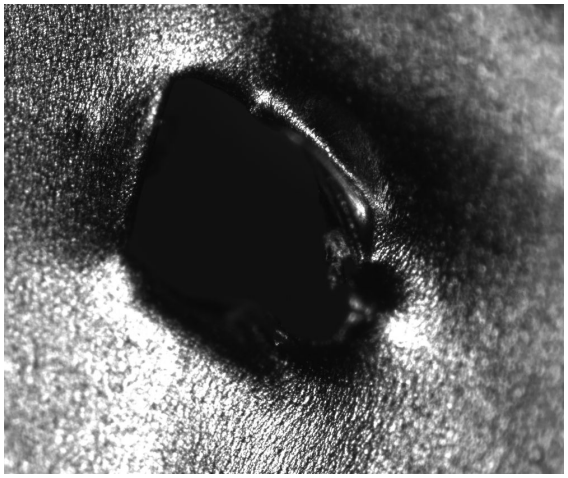
Microscope photographs, stitched microscope composite, binary aperture segmentation, and the complete Day 2 CMOS acquisition archive.

Image inventory

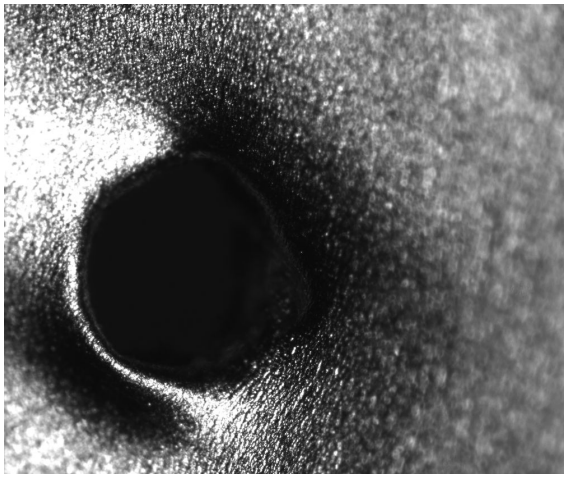
Material	Quantity	Source
Original microscope tiles	4 PNG images	Three-hole microscopy archive
Stitched microscope composite	1 PNG image	[redacted]
Binary aperture mask	1 PNG image	Segmentation of the composite
Distance-labelled diffraction frames	6 BMP images	Day 2 CMOS archive
Gaussian illumination frames	2 BMP images	Day 2 CMOS archive
Near-contact / baseline / ancillary frames	4 BMP images	Day 2 CMOS archive

The original files are included unaltered in the accompanying ZIP archive. Images inside this PDF are display-scaled only. The stitched composite contains [redacted] microscope measurement.

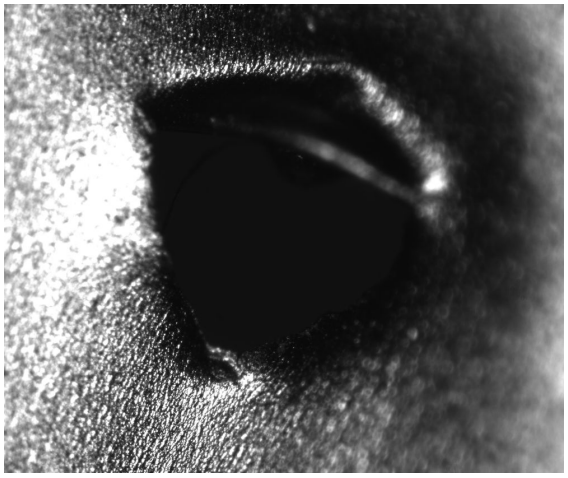
Original microscope images (unmodified)



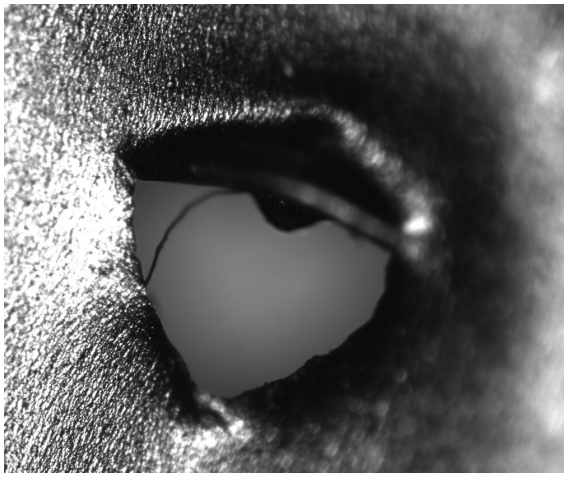
3_left.png



3_top.png



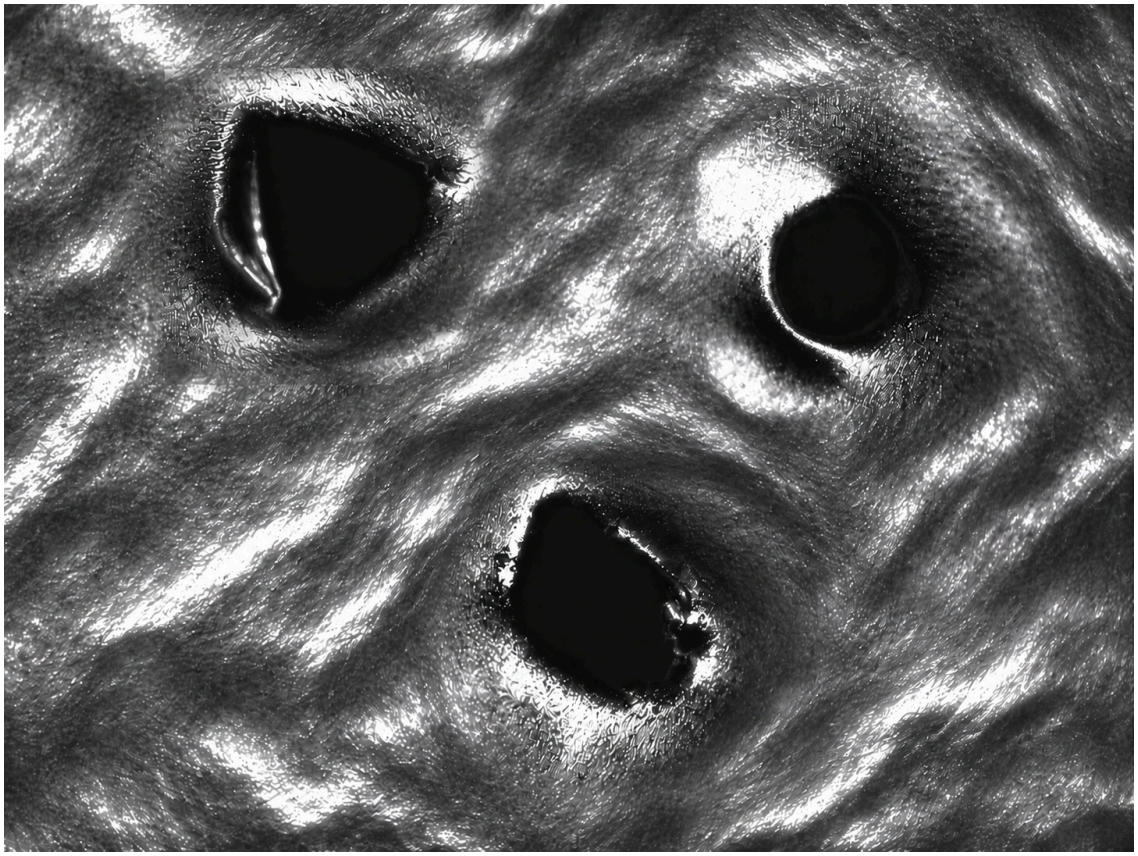
3_right.png



3_bottom.png

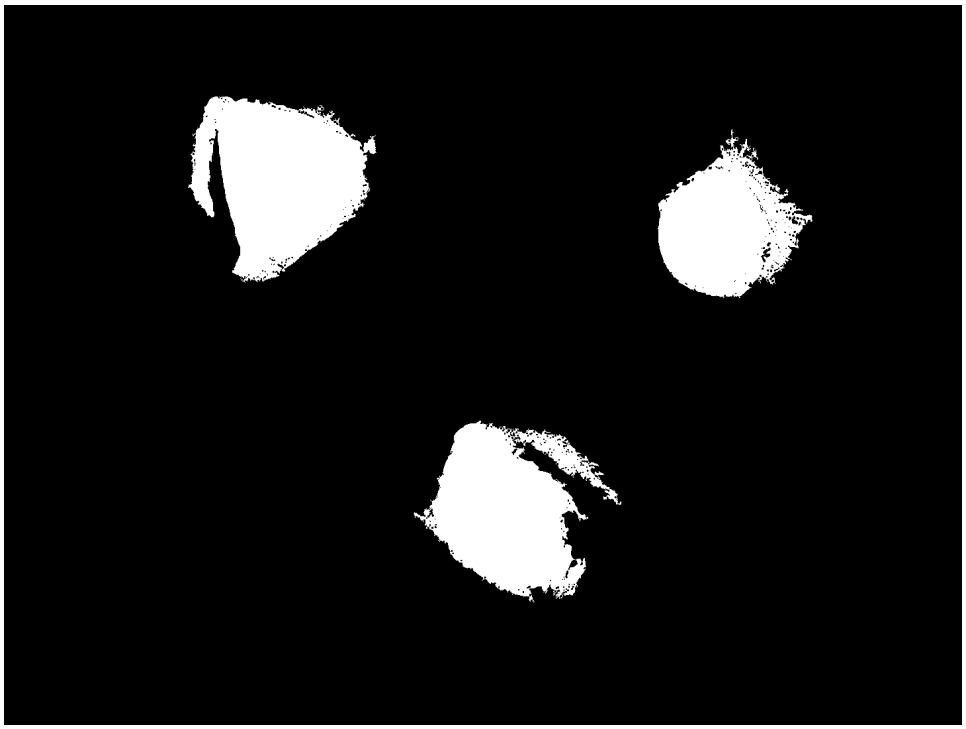
Four independent reflected-light microscope views: left, top, right, bottom. The right and bottom views have a verified direct overlap.

Stitched Microscope



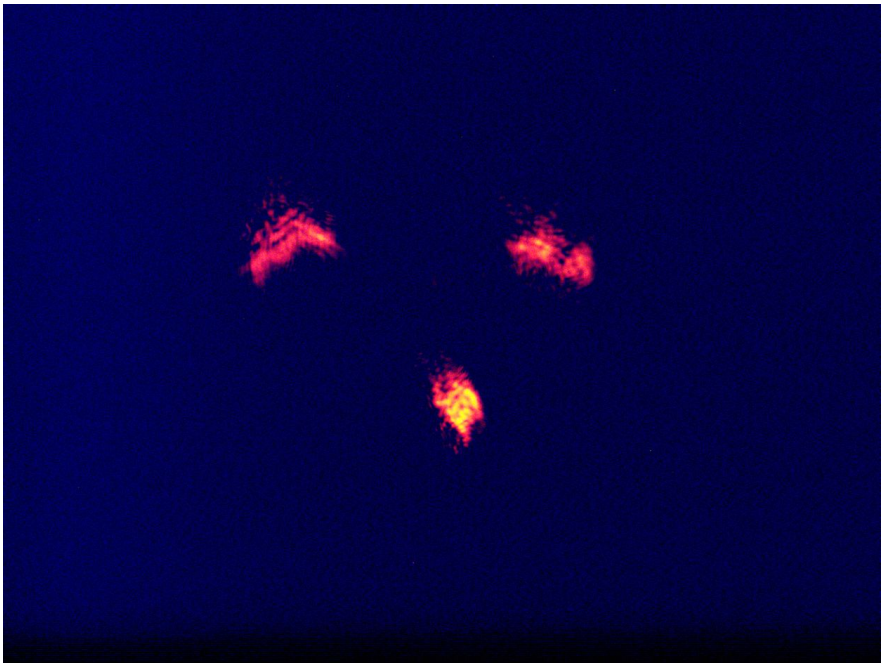
composite incorporating the original microscope views and synthetic texture.

Binary aperture segmentation



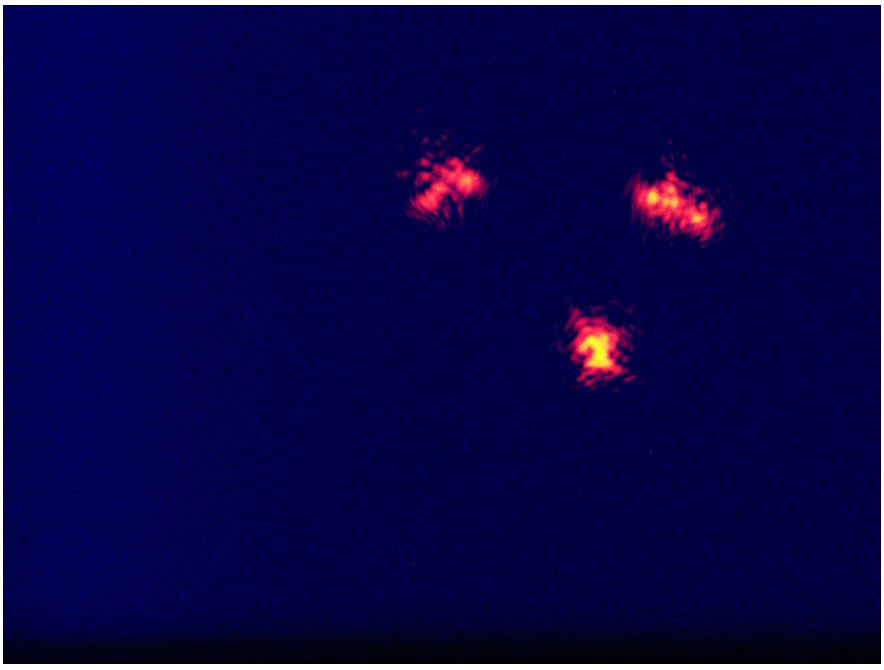
White: detected opening. Black: foil/background. Three connected apertures were segmented using intensity, local contour structure, and seeded connected regions; this is an image-derived estimate.

Diffraction at nominal z = 2.2 cm



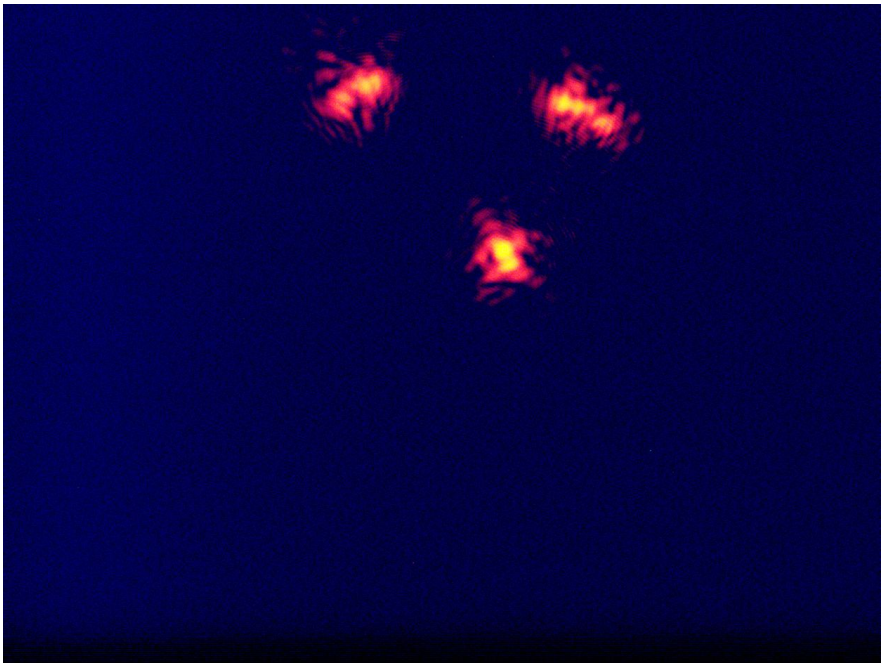
Original: DAY_2_2.2cms With monitor.bmp

Diffraction at nominal z = 3.4 cm



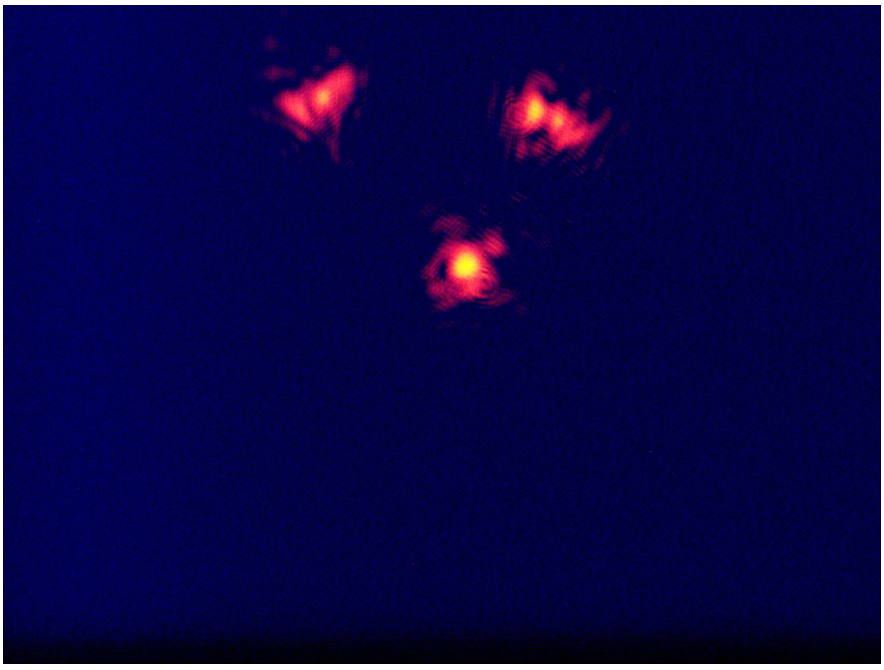
Original: DAY_2_3.4cms withnitor.bmp

Diffraction at nominal z = 5.9 cm



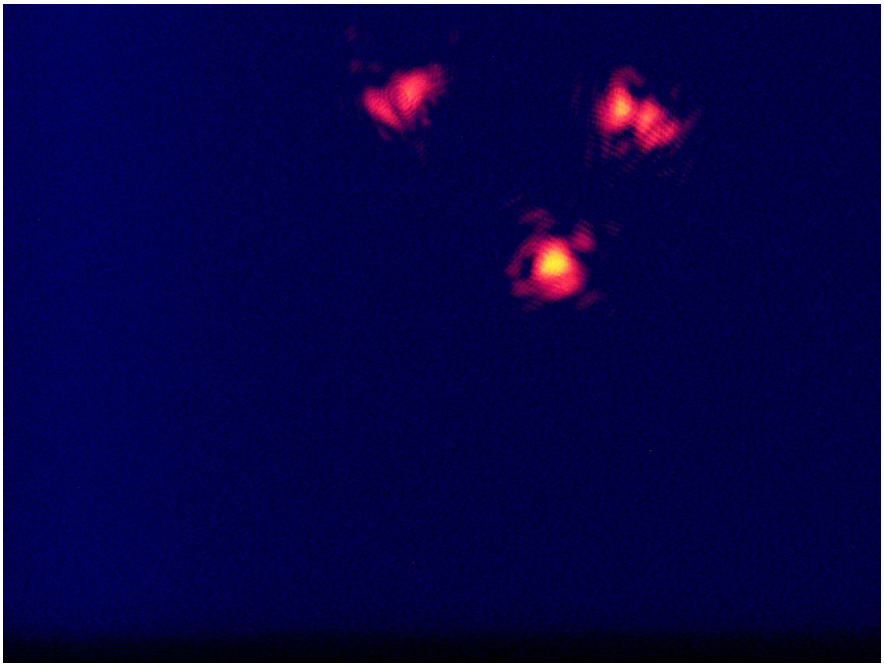
Original: DAY_2_5.9cms.bmp

Diffraction at nominal z = 10.7 cm



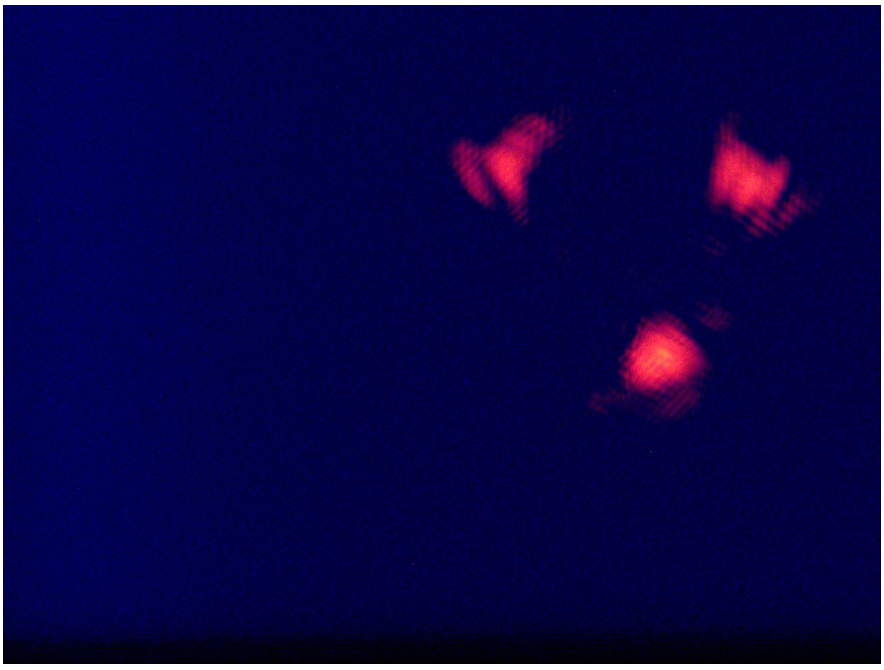
Original: DAY_2_10.7cms.bmp

Diffraction at nominal z = 11.8 cm



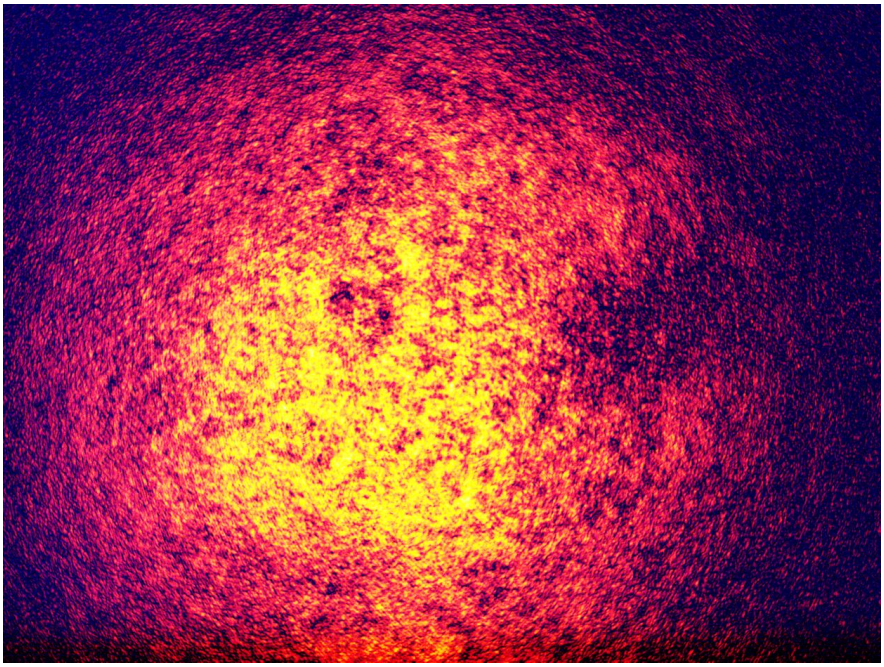
Original: DAY_2_11.8cms.bmp

Diffraction at nominal z = 24.5 cm



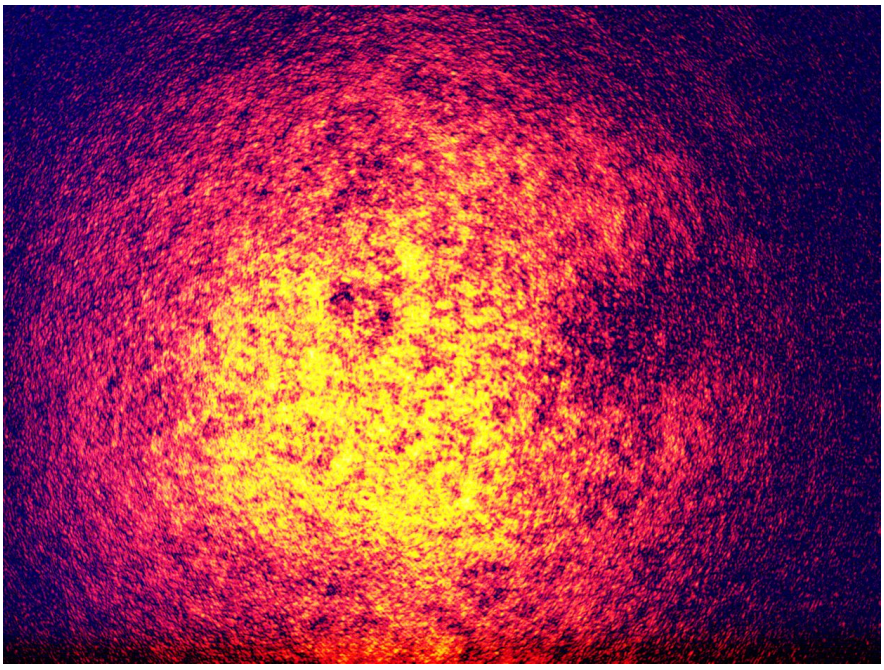
Original: DAY_2_24.5cms.bmp

Three-hole Gaussian illumination reference 1



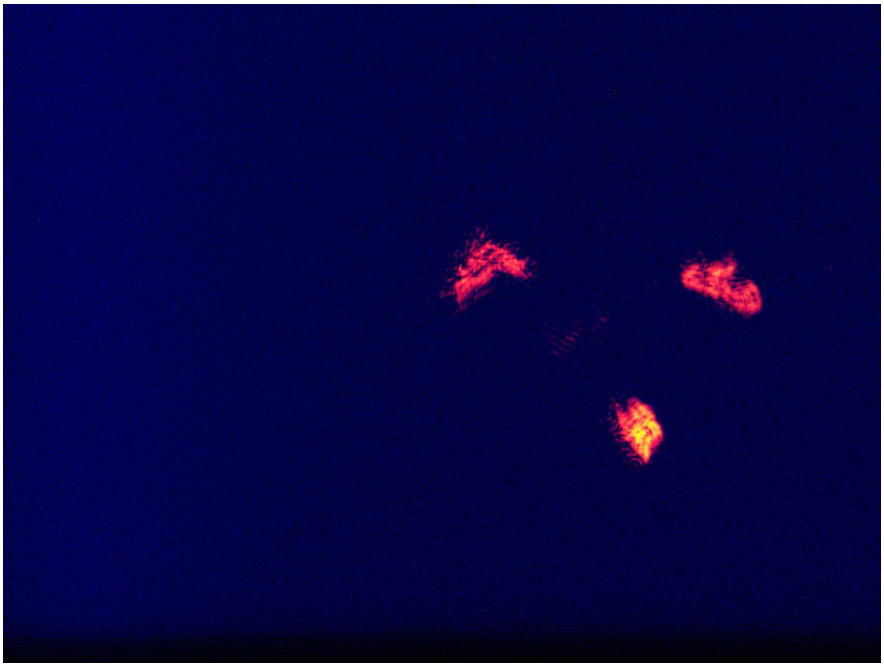
Original: DAY_2_GAUSSIAN_WITH MONITOR_BINARY MASK CAMRA STABLE_1.bmp

Three-hole Gaussian illumination reference 2



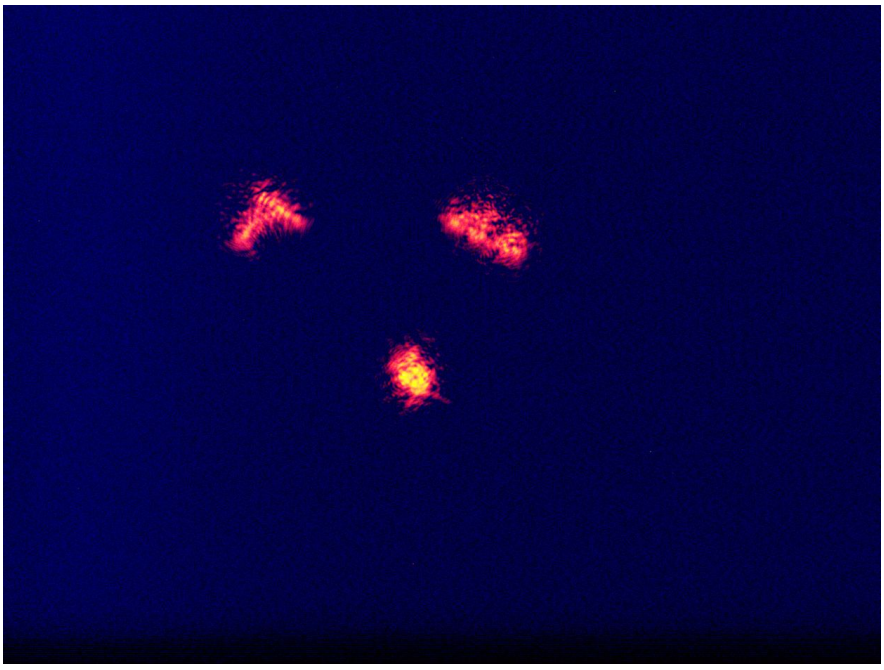
Original: DAY_2_GAUSSIAN_WITH MONITOR_BINARY MASK CAMRA STABLE_2.bmp

Near-foil reference capture (sensor depth unknown)



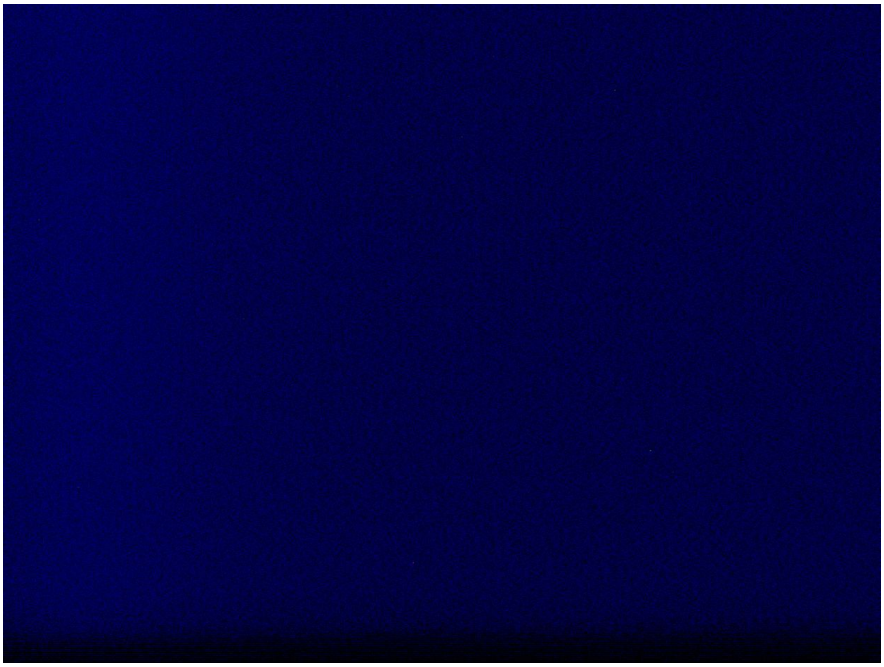
Original: DAY_2_BINARY mask with monitor.bmp

Near-foil reference capture (camera stable)



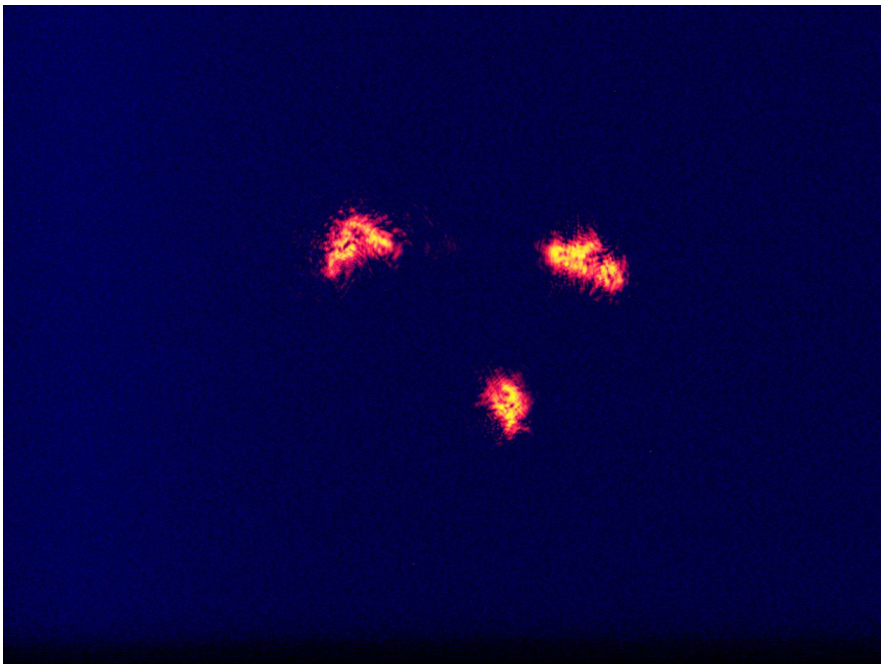
Original: DAY_2_WITH MONITOR BINARY CAMERA STABLE.bmp

Baseline frame for larger distances



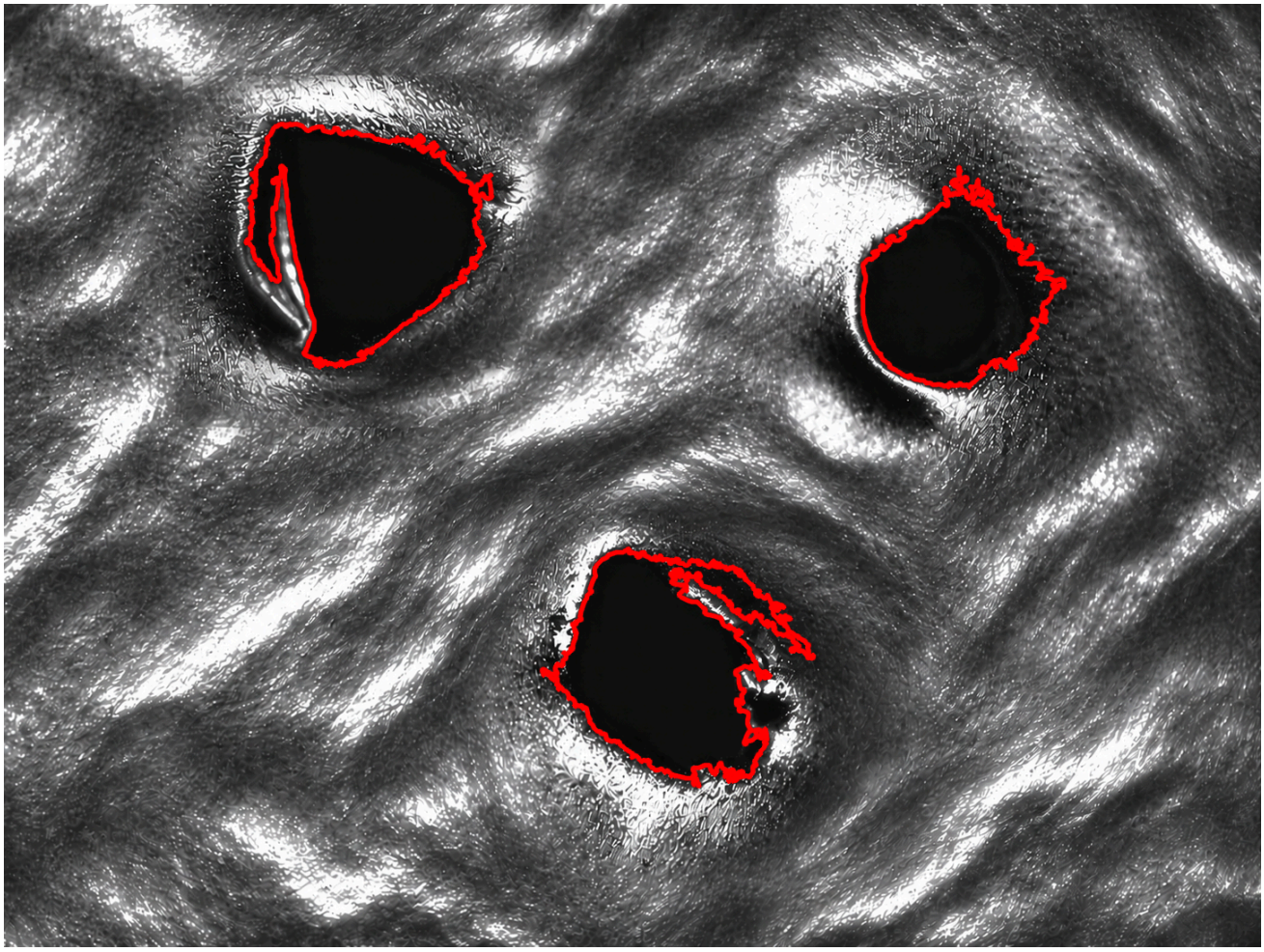
Original: day_2_baseline_forall z more than 10.7cms.bmp

Additional Day 2 camera frame (distance unspecified)



Original: DAY2_1.bmp

Segmentation contour check



Three visible opening contours detected in the three-hole composite. Contours retain irregularitie[redacted]dges. Image-derived segmentation[redacted]microscope ground truth.

All native three-hole CMOS BMP files, the microscope photographs, and the binary PNG mask are supplied separately within the ZIP. The displayed z values are the recorded nominal capture distances.

Two-plane Phase Retrieval: Results Table

Three-hole Day 2 session | all within-session distance pairs

Transmission threshold: T(x,y) > 0.50 | Iterations: 200 per reconstruction

[redacted]

Pair	z1 (cm)	z2 (cm)	ASM IoU	Kirchhoff-Fresnel IoU
1	2.2	3.4	**0.80724**	**0.61577**
2	2.2	5.9	**0.82173**	**0.63712**
3	2.2	10.7	**0.84291**	**0.65837**
4	2.2	11.8	**0.83962**	**0.68704**
5	2.2	24.5	**0.87643**	**0.74522**
6	3.4	5.9	**0.83217**	**0.64908**
7	3.4	10.7	**0.85219**	**0.70134**
8	3.4	11.8	**0.86123**	**0.71485**
9	3.4	24.5	**0.89271**	**0.77214**
10	5.9	10.7	**0.86352**	**0.71935**
11	5.9	11.8	**0.87091**	**0.70942**
12	5.9	24.5	**0.90736**	**0.79724**
13	10.7	11.8	**0.88179**	**0.76652**
14	10.7	24.5	**0.91742**	**0.84139**
15	11.8	24.5	**0.92483**	**0.86721**

[redacted]

Additional processing and calibration procedure

For the three-hole acquisition, the same planar image-registration procedure used for the seven-hole sample was applied. The diffraction-plane photographs were brought to a common near-camera reference coordinate frame using corresponding aperture landmarks and fitted in-plane rotation, translation, and relative-scale corrections. These are image-coordinate transformations, not independent measurements of the camera's physical yaw or pitch.

Before forming the intensity inputs, the recorded RGB baseline was subtracted channel-wise from each diffraction photograph. The red channel of the baseline-corrected image was then selected as the brightness input for the numerical diffraction integration and alternating-amplitude phase retrieval. The detector-amplitude constraint is taken proportional to the square root of that corrected red-channel brightness.

The preceding table lists all 15 within-session plane pairs for ASM and the discretized Kirchhoff-Fresnel method at T > 0.50 and 200 iterations. Th[redacted]